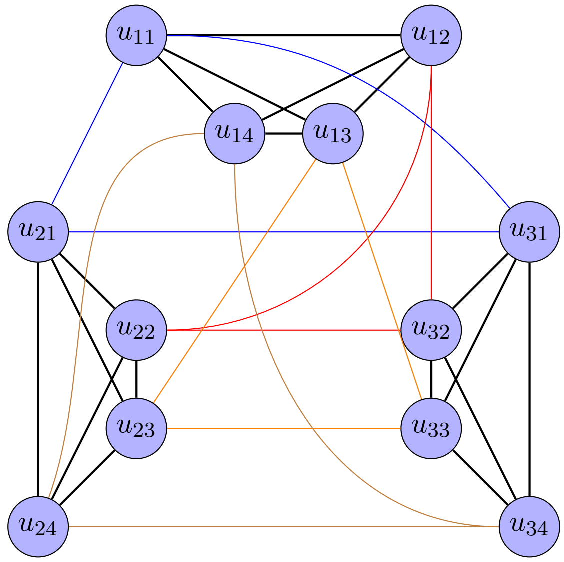
To understand the notation we will use, look at the following example, $K_3 \times K_4$



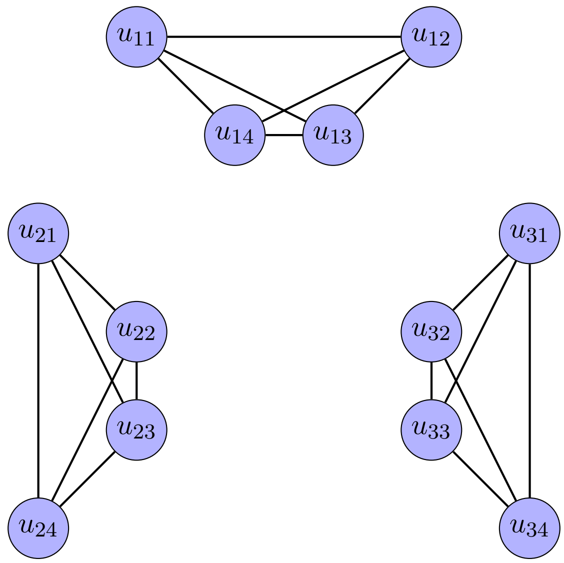
We will denote the vertices $V(K_m \times K_n) := \{u_{ij} : 1 \leq i \leq m, 1 \leq j \leq n\}$ where we have edges:

$$u_{ij}u_{ik}, \quad 1 \leq i \leq m, 1 \leq j, k \leq n, j \neq k \tag{1}$$

$$u_{ij}u_{kj}, \quad 1 \leq i, k \leq m, 1 \leq j \leq n, i \neq k \tag{2}$$

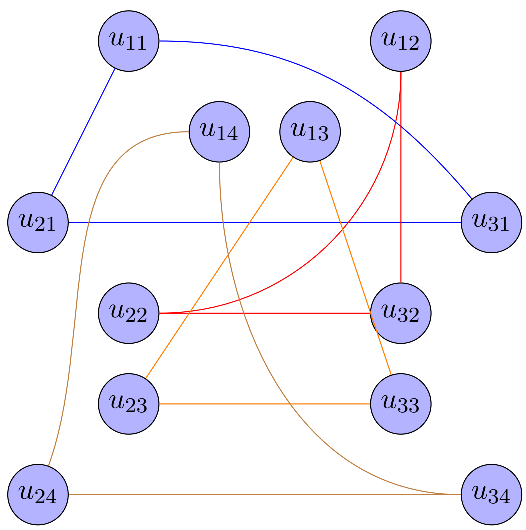
That is, vertices in the same K_n part of the graph share the same first index, and the vertices in the same $K_m x$ part of the graph share the same second index.

Note that $F_1 \cong mK_n$ is the subgraph of G obtained by removing edges described in (2), the K_n subgraphs of G that are disconnected. Graphically, using the example above, F_1 is the face:



A set $S \subseteq V(G)$ is dominating F_1 iff $N[S] = V(F_1)$ in F_1. F_1 consists of m disconnected components isomorphic to K_n. So write $F_1 = \bigcup_{i=1}^{m} C_i$ where each $C_i \cong K_n$ is the component with vertices $\{u_{ij} : 1 \leq j \leq n\}$. In order for S to dominate F_1, S must then contain at least one vertex from each component. Containing a vertex $v \in C_i$ for some $1 \leq i \leq m$ means that every vertex in C_i is in $N[S]$ (since C_i is complete). Note that two vertices are in different components in F_1 if their first index differs. Choosing a vertex from every C_i, $1 \leq i \leq m$, means choosing a set vertices such that every $1, 2, ..., m$ is a first index of some vertex.

We have that $F_2 \cong nK_m$ is the subgraph of G obtained by removing edges described in (1), with $F_2 = \bigcup_{j=1}^{n} D_j$ where each $D_j \cong K_m$ is the component with vertices $\{u_{ij} : 1 \leq i \leq m\}$. Again, using the example above, F_2 is the face:



As explained above, in order for a vertex-set S to dominate F_2, S must contain at least one vertex from each component D_j, and containing a vertex $v \in D_j$, for some $1 \leq j \leq n$, means that every vertex in D_j is in $N[S]$. Note that two vertices are in different components in F_2 if their second index differs. Choosing a vertex from every D_j, $1 \leq j \leq n$, means choosing a set vertices such that every $1, 2, ..., n$ is a second index of some vertex.

So for a set S to be a global dominating set of G with respect to $\{F_1, F_2\}$, S must contain at least one vertex from every C_i and D_j, for $1 \leq i \leq m, 1 \leq j \leq n$. Consider S in the following 2 cases and we will show it is a minimum global dominating set.

1. $m < n$ $\quad S = \{u_{11}, u_{22}, u_{33}, ..., u_{mm}, u_{1,m+1}, ..., u_{1n}\}$
2. $m = n$ $\quad S = \{u_{11}, u_{22}, u_{33}, ..., u_{mm}\}$

In both these cases, this set contains exactly n vertices, one from each D_j, $1 \leq j \leq n$, so it is a minimum. It dominates F_1 and F_2 since it contains a vertex from every component C_i and D_j, $1 \leq i \leq m, 1 \leq j \leq n$. If $m > n$, we use

$$S = \{u_{11}, u_{22}, u_{33}, ..., u_{nn}, u_{n+1,1}, ..., u_{m1}\}$$

This set contains exactly m vertices, one from each C_i, $1 \leq i \leq m$, so it is a minimum. It dominates F_1 and F_2 since it contains a vertex from every component C_i and D_j, $1 \leq i \leq m, 1 \leq j \leq n$.

So $\gamma(K_m \times K_n, \{F_1, F_2\}) = \max\{n, m\}$